

FitBuddy – AI Fitness Plan Generator

Module 7: Technical & System Project Documentation

PROJECT TITLE	CORE TECH	AUTHOR / LEAD	DOC VERSION
FitBuddy AI	Google Gemini AI	Ravi-teja-777	v1.0 (Final)

1.0 Executive Summary & System Overview

FitBuddy is an artificial intelligence-powered health and fitness planning platform built on top of **Google Gemini Models**. The system generates individualized, constraint-aware weekly workout regimens and nutritional recommendations based on user parameters including age, fitness goal, available equipment, and daily schedule availability.

2.0 Tech Stack & Dependencies

Layer	Technology Selected	Purpose in FitBuddy
AI Model	Google Gemini API (Gemini Flash / Pro)	Natural language plan generation & JSON schema structuring
Frontend	HTML5, CSS3, JavaScript / React	Responsive user dashboard and plan display interface
Backend / Middleware	Node.js / Python FastApi	API authentication, prompt formatting, rate limiting
Environment	Dotenv / Secret Manager	Secure storage of API keys and server configuration

3.0 Environment Setup & Installation Guide

Follow these steps to configure and launch the FitBuddy application locally or deploy to server infrastructure:

```
# Step 1: Clone the Repository git clone https://github.com/Ravi-teja-777/FitBuddy.git cd FitBuddy # Step 2: Install Required Dependencies npm install # or pip install -r requirements.txt # Step 3: Set up Environment Variables (.env) GEMINI_API_KEY="your_google_gemini_api_key_here" PORT=3000 # Step 4: Run the Development Server npm start # or python app.py
```

4.0 System Data Flow Architecture

1. Input Capturing

User provides personal profile metrics: Age, Weight, Target Goal (Weight loss/Hypertrophy/Maintenance), Available Equipment, and Session Days.

2. Prompt Assembly

The backend injects user parameters into strict Gemini system instructions enforcing JSON output formats.

3. Gemini API Inference

Google Gemini model processes context, applies safety guardrails, and outputs structured workout splits and dietary macros.

4. Dashboard Rendering

Frontend parses response into interactive daily cards, printable PDF routines, and exercise progress trackers.

5.0 Core Prompt & API Integration Strategy

The core engine relies on a calibrated prompt structure sent to the Gemini API endpoint to ensure consistent output formatting:

```
SYSTEM PROMPT CONTEXT: "You are FitBuddy AI, a certified fitness coach. Generate a structured JSON workout schedule for a user with the following profile: - Goal: {user_goal} | Fitness Level: {user_level} - Equipment: {available_equipment} | Days/Week: {workout_days} Output Requirement: Return strict JSON with keys: 'weekly_split', 'dietary_macros', 'safety_tips'."
```

6.0 Security & Safety Compliance

- API Key Protection:** The Gemini API key is isolated in server-side environment variables and never exposed to the client-side browser.
- Health Safety Guardrails:** System prompts prohibit the generation of extreme caloric restriction plans (< 1200 kcal/day) or high-risk lifting techniques without warnings.
- Data Privacy:** User physical parameters are processed in-memory during plan generation and not stored without user consent.

7.0 Verification & Project Testing Summary

Test Domain	Validation Target	Status
API Response Schema	Ensures valid JSON parsing across various prompt parameters	PASS
Edge Parameters	Validates zero-equipment and home-workout request constraints	PASS
Latency Benchmarking	Response delivery within acceptable UX threshold (< 3.0s)	PASS

8.0 Maintenance & Future Enhancements

Planned Roadmap

- Integration with wearable fitness tracker APIs (Google Fit / Apple Health)
- Computer vision pose-estimation feature for live exercise form checking
- Multilingual fitness plan generation powered by Gemini Multimodal capabilities

